

Matgeo-q5.7.9

AI25BTECH11036-SNEHAMRUDULA

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Q. Question 5.7.9: If

$$A = \begin{bmatrix} -3 & 6 \\ -2 & 4 \end{bmatrix},$$

show that $A^3 = A$.

solution

Given: $\mathbf{A} = \begin{pmatrix} -3 & 6 \\ -2 & 4 \end{pmatrix}$. The characteristic equation is

$$f(\lambda) = |\mathbf{A} - \lambda \mathbf{I}| = \begin{vmatrix} -3 - \lambda & 6 \\ -2 & 4 - \lambda \end{vmatrix} = (-3 - \lambda)(4 - \lambda) - (-12) = \lambda^2 - \lambda - 12$$

(1)

By Cayley–Hamilton, $f(\mathbf{A}) = \mathbf{0}$, hence

$$\mathbf{A}^2 - \mathbf{A} = \mathbf{0} \implies \mathbf{A}^2 = \mathbf{A}.$$

(2)

Multiplying (2) on the right by $\mathbf{A}$ yields

$$\mathbf{A}^3 = \mathbf{A}^2 \mathbf{A} = \mathbf{A} \mathbf{A} = \mathbf{A}^2.$$

(3)

Using (2) in (3),

$$\mathbf{A}^3 = \mathbf{A}.$$

(4)

Graphical Representation

Action of A on vectors (Blue = original, Red = $A \cdot v$)

